

## Practice Questions: Budget Sets

### Question 1. Quantity Discount with Lower Marginal Price

Consider an economy with two goods: apples  $x$  and oranges  $y$ . Oranges have price  $q > 0$ . Apples are sold according to the following pricing rule: the first unit costs  $p > 0$  per apple, but every apple beyond the first costs only  $rp$  per apple, where  $r \in (0, 1)$ . Thus, total spending on apples is

$$C(x) = \begin{cases} px, & 0 \leq x \leq 1, \\ p + rp(x - 1), & x > 1. \end{cases}$$

The consumer has wealth  $w > 0$ . The budget set is

$$B(p, q, w) = \{(x, y) \in \mathbb{R}_+^2 : C(x) + qy \leq w\}.$$

- (a) Draw the budget set. Label all intercepts, slopes, and kink points.
- (b) Is the budget set convex? Explain.
- (c) Suppose preferences are represented by

$$u(x, y) = ax + y, \quad a > 0.$$

Find the demand correspondence  $D(p, q, w)$ .

- (d) Does demand satisfy Walras' law?
- (e) Is demand homogeneous of degree zero in  $(p, q, w)$ ?
- (f) Derive the expenditure function  $e(p, q, U)$ . Is it strictly increasing in  $U$ ?

## Question 2. Buy Two, Get One Free

Apples and oranges are sold at prices  $p > 0$  and  $q > 0$ , respectively. Apples are subject to the following promotion: if a consumer pays for at least 2 apples, the consumer receives one additional apple for free. Goods are perfectly divisible. The consumer's wealth is  $w > 0$ .

Let  $x$  denote apples available for consumption after the promotion.

- (a) Write the minimum cost  $C(x)$  of consuming  $x$  apples. *Hint: Think carefully about the region  $2 \leq x \leq 3$ .*
- (b) Draw the budget set

$$B(p, q, w) = \{(x, y) \in \mathbb{R}_+^2 : C(x) + qy \leq w\}.$$

Label the flat segment, if any.

- (c) Suppose preferences are represented by

$$u(x, y) = 2x + y.$$

Find the demand correspondence  $D((p, q), w)$ .

- (d) Does the consumer always spend all wealth?
- (e) Derive the expenditure function  $e((p, q), U)$ . Is it continuous in  $U$ ? Is it strictly increasing in  $U$ ?

### Question 3. Apple Membership / Two-Part Tariff

A store offers two ways to buy apples. A consumer can either buy apples at the regular unit price  $p > 0$ , or pay a membership fee  $F > 0$  and then buy apples at the discounted unit price  $rp$ , where  $r \in (0, 1)$ . Oranges have price  $q > 0$ . The consumer's wealth is  $w > 0$ .

The total cost of consuming  $x$  apples is

$$C(x) = \min\{px, F + rpx\}.$$

- (a) Draw the budget set. Label the switching point

$$\bar{x} = \frac{F}{p(1-r)}.$$

- (b) Is the budget set convex? Explain geometrically.

- (c) Let preferences be represented by

$$u(x, y) = x + \beta y, \quad \beta > 0.$$

Find the demand correspondence.

- (d) For what values of  $(p, q, F, r, \beta)$  is the consumer indifferent between the regular pricing option and the membership option?
- (e) Is the demand homogeneous of degree zero in  $(p, q, w)$ ? What if  $F$  is also multiplied by  $\alpha > 0$ ?

### Question 4. Apple Voucher / Subsidy for First Units

The consumer has wealth  $w > 0$ . Oranges have price  $q > 0$ . Apples are sold with a subsidy: the first  $\bar{x} > 0$  apples cost  $p_L > 0$  each, and all apples above  $\bar{x}$  cost  $p_H > p_L$  each. Thus,

$$C(x) = p_L \min\{x, \bar{x}\} + p_H \max\{x - \bar{x}, 0\}.$$

- (a) Draw the budget set. Label the kink and the two slopes.
- (b) Is the budget set convex?
- (c) Suppose preferences are represented by

$$u(x, y) = ax + y.$$

Find  $D(p_L, p_H, q, w)$ .

- (d) Does demand satisfy Walras' law?
- (e) Is demand homogeneous of degree zero in  $(p_L, p_H, q, w)$ ?
- (f) Compute the expenditure function  $e(p_L, p_H, q, U)$ .

## Question 5. Quantity Rationing

Apples and oranges have prices  $p > 0$  and  $q > 0$ . However, because of rationing, the consumer cannot buy more than  $\bar{x} > 0$  apples. The consumer has wealth  $w > 0$ .

$$B(p, q, w) = \{(x, y) \in \mathbb{R}_+^2 : px + qy \leq w, x \leq \bar{x}\}.$$

- (a) Draw the budget set for the cases  $w/p < \bar{x}$  and  $w/p \geq \bar{x}$ .
- (b) Suppose preferences are represented by

$$u(x, y) = ax + y.$$

Find the demand correspondence.

- (c) Does the consumer spend all wealth?
- (d) Is the demand homogeneous of degree zero in  $(p, q, w)$ ?
- (e) Derive  $e((p, q), U)$ . Does the rationing constraint ever bind in the expenditure minimization problem?

## Question 6. Nonlinear Total Cost: Square-Root Discount

Oranges have price  $q > 0$ . Apples have a nonlinear total cost

$$C(x) = p\sqrt{x}, \quad x \geq 0,$$

where  $p > 0$ . The consumer has wealth  $w > 0$ .

- (a) Draw the budget set

$$B(p, q, w) = \{(x, y) \in \mathbb{R}_+^2 : p\sqrt{x} + qy \leq w\}.$$

- (b) Is the budget set convex?

- (c) Suppose preferences are represented by

$$u(x, y) = x + ay, \quad a > 0.$$

Find  $D(p, q, w)$ .

- (d) Does the demand correspondence satisfy Walras' law?

- (e) Is  $D$  homogeneous of degree zero?

- (f) Compute  $e(p, q, U)$ .

## Question 7. Fixed Package Promotion

Apples and oranges are sold at prices  $p > 0$  and  $q > 0$ . In addition, the store offers a package containing one apple and one orange for total price  $s > 0$ . The consumer may buy any nonnegative real number of packages, and goods are perfectly divisible.

Let  $x$  be apples consumed and  $y$  oranges consumed.

- (a) Write the consumer's budget set.

*Hint: The consumer can combine packages with separate purchases of apples and oranges.*

- (b) Draw the budget set for the cases

$$s < p + q, \quad s = p + q, \quad s > p + q.$$

- (c) Suppose preferences are represented by

$$u(x, y) = \alpha x + \beta y.$$

Find the demand correspondence.

- (d) Does the demand satisfy Walras' law?
- (e) Is the demand homogeneous of degree zero in  $(p, q, s, w)$ ?

## Question 8. Coupon with Dollar Rebate

The consumer has wealth  $w > 0$ . Apples have price  $p > 0$ , oranges have price  $q > 0$ . The consumer receives a coupon that pays for the first  $S > 0$  dollars of apple purchases. Thus, the out-of-pocket cost of consuming  $x$  apples is

$$C(x) = \max\{px - S, 0\}.$$

- (a) Draw the budget set. Label the horizontal/free segment.
- (b) Is the budget set convex?
- (c) Suppose preferences are represented by

$$u(x, y) = x + ay.$$

Find  $D(p, q, w, S)$ .

- (d) Does the consumer necessarily spend all wealth?
- (e) Is demand homogeneous of degree zero in  $(p, q, w)$ ? What changes if  $S$  is also scaled by the same factor?
- (f) Derive the expenditure function  $e(p, q, S, U)$ . Is it strictly increasing in  $U$ ?